ZINC SULFATE

Diarrhoea in children: 10–14 days

Take with food

CHEMICAL NAME $ZnSO_4 \cdot H_2O$ / $7H_2O$ (zinc sulfate monohydrate/heptahydrate)
CLASS Trace-element mineral supplement
TRADE NAMES Solvazinc · Zincat · Z-Span · Zinc sulfate dispersible (WHO)

THERAPEUTIC USES / INDICATIONS

- Acute diarrhoea in children (with ORS; WHO/UNICEF).
- Zinc deficiency (malnutrition, malabsorption, parenteral nutrition, acrodermatitis enteropathica).
- Wilson's disease (maintenance, zinc acetate).
- Wound healing support in deficiency.

PHARMACODYNAMICS

- Restores intestinal epithelium and brush-border enzymes; enhances immunity → shorter, less severe diarrhoea and fewer recurrences.
- Cofactor for >300 enzymes.

ADMINISTRATION

- **PO** dispersible tablets dissolved in water or breast milk.
- Take with or after food to reduce nausea.
- Doses stated as elemental zinc.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (empty stomach).
- Metallic taste.
- Abdominal discomfort.

ADVERSE EFFECTS (SERIOUS)

- Copper deficiency (anaemia, neutropenia, neuropathy) with prolonged high doses.
- Gastric irritation.

DOSAGES

- **Diarrhoea, child <6 months:** 10 mg elemental zinc daily for 10–14 days.
- **Diarrhoea, child ≥6 months:** 20 mg daily for 10–14 days.
- **Deficiency (adult):** 25–50 mg elemental zinc 1–3 times daily.
- 220 mg zinc sulfate ≈ 50 mg elemental zinc.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** renal failure (accumulation).

KEY INTERACTIONS

- Reduces absorption of tetracyclines, fluoroquinolones, penicillamine; separate by 2–3 h.
- **Iron, calcium:** ↓ zinc absorption.

PHARMACOKINETICS

- Absorbed in small intestine (20–30%); ↓ by phytates, iron, calcium.
- Excreted mainly in faeces.

NURSING MANAGEMENT

- Teach caregivers to complete full 10–14-day course even after diarrhoea stops.
- Give with ORS and continued feeding.
- Monitor for vomiting after dose (repeat if within 30 min).

PATIENT EDUCATION

- Dissolve the tablet in a little water or breast milk.
- Continue for 10–14 days to protect against future diarrhoea.
- Give after food if it causes vomiting.

KEY POINTS

- Zinc + ORS for childhood diarrhoea.
- 10 mg (<6 mo) / 20 mg (≥6 mo) for 10–14 days.
- Nausea — give with food.
- Long-term high doses → copper deficiency.